

Advanced SQL

Null Functions

Practical SQL Worksheet

Question 1

Table 1: Employees

Employee id	Name	Salary with default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2

Employee id	Name	Department name
1	Alice	HR
2	Bob	IT
3	Charlie	NOT Assigned
4	Dana	Finance

Question 3: Orders

a. Finding Customers orders with Null Customer id using IS Null

SELECT * FROM Orders WHERE Customer id IS Null;

Order id	Customer id
103	Null

Question 4

Showing all order and where delivery is Null mark as pending.

Order id	Customer id	Delivery date
101	201	2024-12-01
102	202	Pending
103	Null	2024-12-03

Question 5: Students

Student id	Name	Final grade
1	Ethan	85
2	Maya	0
3	Olivia	90

~~Select * from students~~
 Select final grade,
 IF NULL (grade, '0') AS
 updated grade/final grade.
 from students;

Question 6:

Not graded Count
1

~~Select student_id,~~
 from students
 where grade IS NULL;

Question 7: Products

Product id	Name	Final price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Select Product id,
 Name,
 Price IF NULL (discount
 0)
 AS final price
 from products;

Question 8: Table 5.

Customers
Missing_email Count
2

Select COUNT (*) AS missing_email Count,
 from Customers
 where email IS NULL;

Question 9

Customer id	Name	email
1	Linda	No email
2	Joseph	Joseph@mail.com
3	Mig	No email

Select Customer id,
 Name,
 IF NULL (email, 'no email') AS
 email display
 from Customers

Question 10: Payments

Table 6: Payments

Payment id	Method display	Status
302		
301		
303	Unknown	Failed

Select Payment id,
If null (method, 'unknown')
As method display, status
From Payments;

Question 11:

Table 7: Inventory

Item id	Item name	quantity	Unit
1	Pen	0	
2	Notebook	150	
3	Eraser	0	

Select Item id, item name,
(COALESCE(quantity, 0)) AS
quantity checked
From Inventory;

Question 12:

Table 8: Employee Extra

Emp id	First Avail reward	300	100	0
1				
2				
3				

Select emp-id,
COALESCE(bonus, commission)
As first available reward
From Employee Extra;

Question 13

Table 9: Classes

no room count	2
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Select Count(*) AS no room count
from Classes
Where room IS Null;

Question 14.

Table 10: Attendance.

Student id	Date	Attendance Status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

SELECT Student id, date,
 COALESCE (Status, 'not marked') AS attendance Status
 FROM Attendance;

Question 15

Table 11: Bank Accounts

account-id	type-display	balance	Account id,
A1	Savings	00	COALESCE (account type, 'unknown')
A2	Current	5000	AS type display)
A3	Unknown	2000	If NULL (balance, 00) AS

balance checked
 FROM Bank Accounts;

Question 16.

Table 12: Projects

Project-id	title	Start date display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Start date display

SELECT project-id, title,

COALESCE (CAST (start_date AS CHAR), 'TBD') AS

Start display
 FROM Projects;

Question 17

Table 13: Reviews

review-id	Product-id	Comment-display	Rating-display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Worst fine	3

Question 17 Syntax

SELECT review-id, product-id,

COALSC (comment, no comment) AS commentdisplay,

IFNULL (rating, 0) AS rating-display,

FROM reviews;

Question 18

Table 14: ~~User Settings~~ Suppliers

Supplier-id	name	contact-number
1	Global goods	123456789
2	Best Suppliers	987654321
3	Value Source	No Contact

SELECT Supplier-id, name,
COALSC (phone, alt-phone, nocontact)
AS contact-number,
FROM Suppliers;

Question 19

Table 15: User Settings

user-id	theme-set	language-set	timezone-set
1	light	English	UTC
2	Dark	English	UTC +1
3	light	English	UTC

SELECT user-id,

COALSC (theme, 'light') AS theme-set,

COALSC (language, 'English') AS language-set,

IFNULL (timezone, 'UTC') AS timezone-set

FROM User-Settings;

Question 20

Table 16: Maintenance

record-id	Machine-id	Issue-log	Technician-name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Trimmed	Alex

Question 20 Synthes.
Select record id, machine-id,
COMESCE (issue, 'unknown issue') / AS issue_log,
COMESCE (technician, 'not Assigned') / AS technician
name
FROM Maintenance;